STAT 405 Project Proposal

Team Member:

- Haoyu You (Student ID: 34735811)
- Ruohan Sun (Student ID: 59673772)

Github website:

https://github.com/HaoyuYou/Bayesian-Statistics-Project.git

Dataset

main:

The dataset used in this project is the Bike Sharing Dataset from the UCI Machine Learning Repository. It contains information about daily bike rental usage in a bike sharing system from 2011 to 2012.

We use the daily dataset (`day.csv`), which includes 731 observations, where each observation represents one day. The dataset records the total number of bike rentals (`cnt`) along with several explanatory variables related to time and weather conditions.

Key variables include temperature (`temp`), humidity (`hum`), wind speed (`windspeed`), season (`season`), weather situation (`weathersit`), and whether the day is a working day (`workingday`). These variables help explain patterns in bike rental demand.

```
data <- read.csv("data/day.csv")
head(data)
```

```
##   instant     dteday season yr mnth holiday weekday workingday weathersit
## 1       1 2011-01-01      1  0    1       0       6          0          2
## 2       2 2011-01-02      1  0    1       0       0          0          2
## 3       3 2011-01-03      1  0    1       0       1          1          1
## 4       4 2011-01-04      1  0    1       0       2          1          1
## 5       5 2011-01-05      1  0    1       0       3          1          1
## 6       6 2011-01-06      1  0    1       0       4          1          1
##       temp    atemp      hum windspeed casual registered  cnt
## 1 0.344167 0.363625 0.805833 0.1604460    331        654  985
## 2 0.363478 0.353739 0.696087 0.2485390    131        670  801
## 3 0.196364 0.189405 0.437273 0.2483090    120       1229 1349
## 4 0.200000 0.212122 0.590435 0.1602960    108       1454 1562
## 5 0.226957 0.229270 0.436957 0.1869000     82       1518 1600
## 6 0.204348 0.233209 0.518261 0.0895652     88       1518 1606
```

URL:

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

backup:

Scientific Question:

How do weather conditions and seasonal factors influence daily bike rental demand? Does a more flexible Bayesian count model provide a better fit and more realistic uncertainty quantification than a simpler baseline model?

More specifically, we aim to model how variables such as temperature, humidity, wind speed, season, and working day status affect the total number of bike rentals (`cnt`).

Bike sharing demand is influenced by both environmental conditions and human behavior patterns. Understanding these relationships can help improve demand prediction and support better planning for urban transportation systems.

Using Bayesian modelling allows us to quantify uncertainty in parameter estimates and make probabilistic interpretations of the effects of different predictors on bike rental demand.

Methodology:

We plan to build Bayesian models to explain and predict the number of daily bike rentals. In particular, we will consider two models: a simple baseline model and a more flexible regression model incorporating environmental predictors.

Before fitting the Bayesian models, we will perform exploratory data analysis (EDA) to understand the distributions of key variables and their relationships with bike rental demand. This includes visualizing the distribution of the response variable (`cnt`) and examining how it varies with weather conditions and seasonal factors.

Model 1: Naive Poisson Model

As a baseline, we first consider a simple Poisson model where all days share the same rental rate:

$$cnt_i \sim \text{Poisson}(\lambda)$$

This model assumes that the expected number of rentals is constant across all observations and does not depend on any explanatory variables. We place a prior distribution on the rate parameter:

$$\lambda \sim \text{Gamma}(1, 1)$$

This naive model provides a simple reference point against which more complex models can be compared.

Model 2: Bayesian Poisson Regression Model

To capture the influence of environmental and temporal factors, we then consider a Poisson regression model where the rental rate varies across observations:

$$cnt_i \sim \text{Poisson}(\lambda_i)$$

with

$$\log(\lambda_i) = \beta_0 + \beta_1 temp_i + \beta_2 hum_i + \beta_3 windspeed_i + \beta_4 workingday_i + \beta_5 season_i$$

We place prior distributions on the regression coefficients:

$$\beta_j \sim \text{Normal}(0, 10^2)$$

This model allows us to quantify how weather conditions and seasonal factors influence bike rental demand.

Inference

The models will be implemented using the simPPLe probabilistic programming framework used in this course. Posterior inference will be performed using MCMC sampling.

We will estimate:

- posterior distributions of model parameters
- uncertainty in predicted bike rental demand
- the relative importance of different predictors

Finally, we will compare the two models using posterior predictive checks to evaluate their ability to explain the observed data.

Contribution plan:

This is a team project and both team members will contribute to all stages of the project.

Ruohan Sun

- Data preprocessing and exploratory data analysis (EDA)
- Implementation of the Bayesian models
- Posterior inference and model diagnostics
- Visualization of results

Haoyu You

- Literature review and background research
- Model formulation and prior specification
- Model comparison and interpretation of results
- Writing and editing the final report